LONG-PERIOD COMET

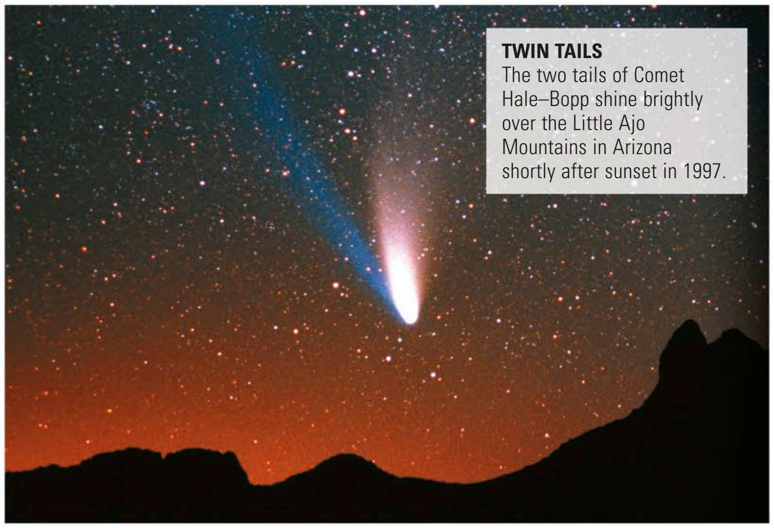
Hale–Bopp

CLOSEST APPROACH TO THE SUN 85 million miles (137 million km)

ORBITAL PERIOD 2,530 and 4,200 years

FIRST RECORDED July 23, 1995

Comet Hale–Bopp was discovered independently and accidentally by the American amateur astronomers Alan Hale and Thomas Bopp, who were looking at Messier objects in the clear skies of the western US (it was close to M70). Later, after the orbit had been calculated, Hale–Bopp was found to be at a distance of over 600 million miles (1 billion km). This is between the orbits of Jupiter and Saturn and is an almost unprecedented distance for the discovery of a nonperiodic comet. The orbit showed that it had been to the inner solar system before, some 4,200 years ago, but because it passed close to Jupiter a few months after discovery, it will return again in about 2,510 years. Hale–Bopp passed perihelion on April 1, 1997. It was one of the brightest comets of the century—not, like Hyakutake, because it came very close to Earth, but simply because it had a huge nucleus, about 22 miles (35 km) across.



TWIN TAILS
The two tails of Comet Hale–Bopp shine brightly over the Little Ajo Mountains in Arizona shortly after sunset in 1997.

SHORT-PERIOD COMET

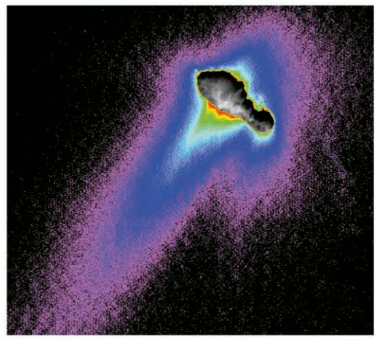
Borrelly

CLOSEST APPROACH TO THE SUN 126 million miles (203 million km)

ORBITAL PERIOD 6.86 years

FIRST RECORDED December 28, 1904

The flyby of NASA's Deep Space 1 mission on September 22, 2001, revealed that this periodic comet has a nucleus shaped like a bowling pin, about 5 miles (8 km) long. Reflecting on average only 3 percent of the sunlight that hits it, Borrelly has the darkest known surface in the inner solar system. Any ice in the nucleus is hidden below the hot and dry, mottled, sooty black surface.



DEEP SPACE 1 IMAGE
The production of the jets of gas and dust emanating from Borrelly's nucleus is eroding the surface. There is a possibility that the nucleus will split in two in the future.

SHORT-PERIOD COMET

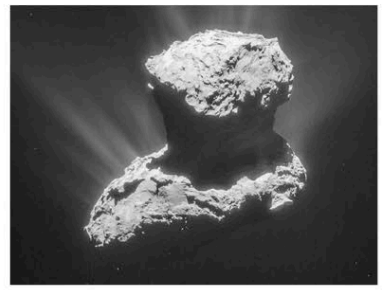
67P/Churyomov–Gerasimenko

CLOSEST APPROACH TO THE SUN 322 million miles (518 million km)

ORBITAL PERIOD 6.44 years

FIRST RECORDED September 11, 1969

This was the target of ESA's Rosetta mission, which orbited it from September 2014 until September 2016, and deployed a partially successful lander (Philae) to its surface in November 2014. The twin lobed shape of the nucleus suggests that it grew by a gentle collisional merger between two bodies. Rugged areas are icy "bedrock," whereas smoother areas are dusty, with some icy boulders.



COMET'S ATMOSPHERE
This Rosetta image, 5 months before perihelion, shows jets of gas and dust escaping from the comet's surface as it became warmed by the Sun.

SHORT-PERIOD COMET

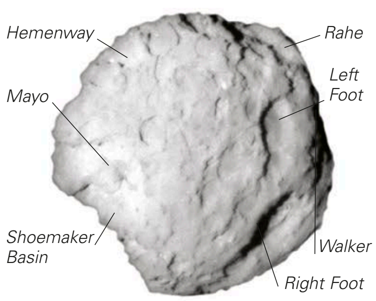
Wild 2

CLOSEST APPROACH TO THE SUN 147 million miles (236 million km)

ORBITAL PERIOD 6.39 years

FIRST RECORDED January 6, 1978

Wild 2 is a relatively fresh comet that was brought into an orbit in the inner solar system as recently as September 1974, when it had a close encounter with Jupiter. It is too faint to be seen with the naked eye, since its nucleus is only 3.4 miles (5.5 km) long. Wild 2's present path around the Sun takes it very close to the orbits of both Mars and Jupiter. It may oscillate between its present orbit and an orbit with a period of about 30 years that brings it only as close as Jupiter. Wild 2 was chosen for NASA's Stardust mission (see panel, below) because the spacecraft could fly by at the relatively low speed of 13,600 mph (21,900 kph), capturing comet dust on the way.



CLOSE-UP OF NUCLEUS
The surface of the nucleus is covered by steep-walled depressions hundreds of yards (meters) deep. They are mostly named after famous cometary scientists.

EXPLORING SPACE

THE STARDUST MISSION

The Stardust spacecraft flew by Wild 2 on January 2, 2004. It has captured both interstellar dust and dust blown away from the comet's nucleus. Aerogel placed on an extended tennis-racket-shaped collector was used to capture the particles without heating them up or changing their physical characteristics. The craft returned to Earth in 2006, and the collector, stowed in a canister, parachuted to safety in the desert in Utah.

AEROGEL
Although it has a ghostly appearance, aerogel is solid. It is a silicon-based sponge-like foam, 1,000 times less dense than glass.

SHORT-PERIOD COMET

Shoemaker–Levy 9

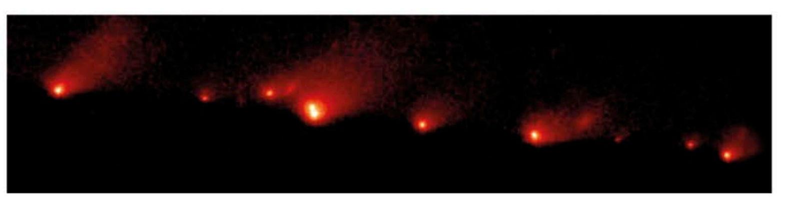
ORBITAL DISTANCE FROM JUPITER 56,000 miles (90,000 km)

ORBITAL PERIOD AROUND JUPITER 2.03 years

FIRST RECORDED March 25, 1993

Unlike normal comets, this one was discovered in orbit around Jupiter by the American astronomers Gene and Carolyn Shoemaker and David Levy. Even more remarkably, it was in 22 pieces, having been ripped apart on July 7, 1992, when it passed close to Jupiter. These fragments subsequently crashed into the atmosphere in Jupiter's southern hemisphere in July 1994 (see p.181). Observatories all over the world and the Hubble Space Telescope witnessed the sequence of events. The nucleus was originally just over 0.6 miles (1 km) across and was most likely captured by Jupiter in the 1920s.

SHATTERED NUCLEUS
This false-color Hubble Space Telescope image show strung-out fragments of the nucleus, each surrounded by its own coma, 2 months before they impacted onto Jupiter.



CAROLYN SHOEMAKER

After taking up astronomy at the age of 51 after her three children had grown up, Carolyn Shoemaker (b. 1929) has now discovered over 800 asteroids and 32 comets. She uses the 18-in (46-cm) Schmidt wide-angle telescope at the Palomar Observatory in California. Her patience and attention to detail are vital when it comes to inspecting photographic plates that are taken about an hour apart and then studied stereoscopically. Typically, 100 hours of searching are required for each comet discovery. Carolyn was married to Gene Shoemaker (see p.139).